Use Case Diagram and Specifications - Anime World Shop

Related documents: Requirement Analysis - ERD - API documentation

The diagrams are written in PlantUML (`diagrams/*.puml`) and exported as PNG. To change one, edit the `.puml` file and run:

```
java -jar plantuml.jar -tpng -Sdpi=150 docs/requirements/diagrams/use-case-*.puml
```

The Word copies (`.docx`) are made from these Markdown files with `python make_docx.py docs/requirements` (needs `pip install pypandoc_binary python-docx`); the PDFs are exported from Word.

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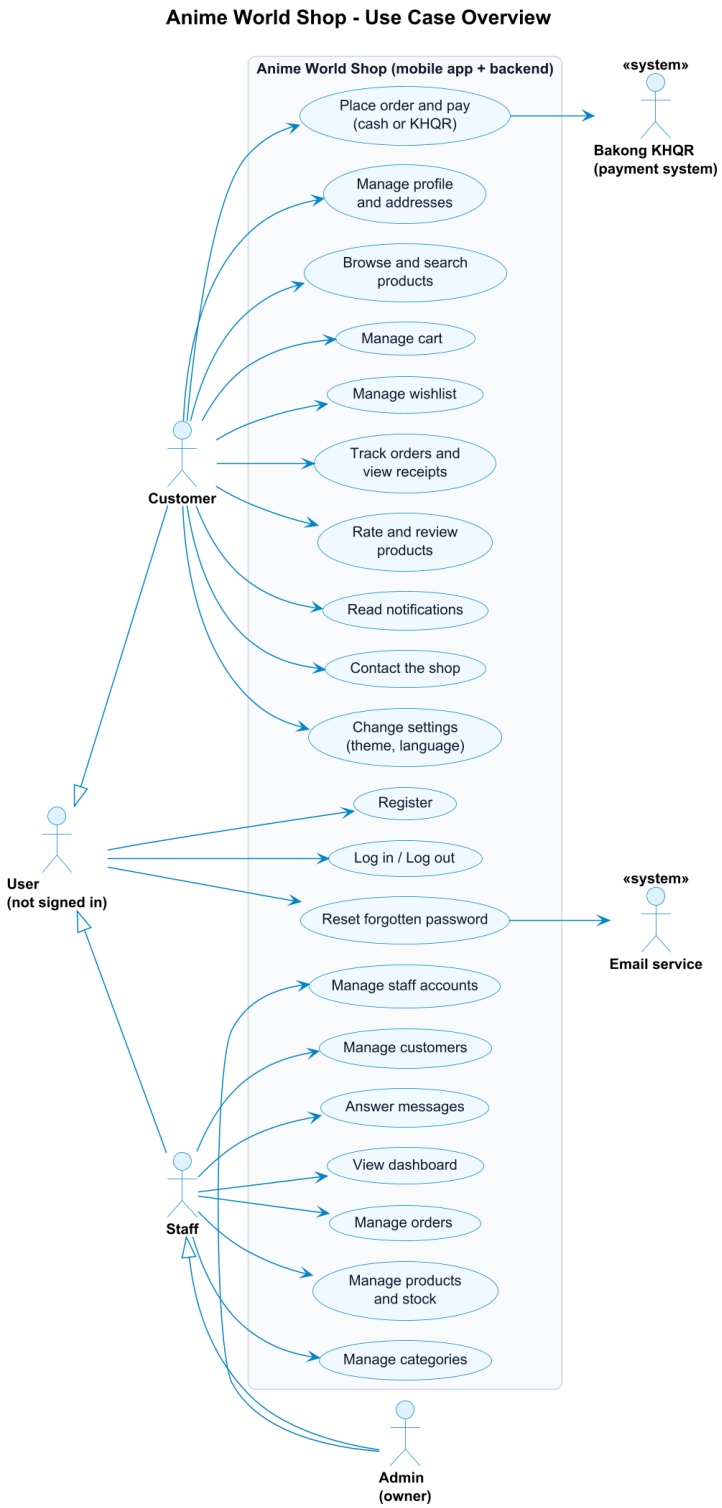
1. Actors

Actor	Type	Description
User (not signed in)	Primary	Anyone who opens the app without being logged in. Can only register, log in or reset a password.
Customer	Primary	A registered shopper. A specialisation of User: everything a User can do, plus shopping.
Staff	Primary	Shop staff. A specialisation of User. Uses the admin tabs of the app or the React web admin at `/dashboard` for the day-to-day work (orders, products, categories, messages, reviews, viewing customers).
Admin	Primary	The shop owner. A specialisation of Staff: everything staff do, plus managing staff accounts, promo codes and blocking customers. Also uses the panel at `/admin`.
Bakong KHQR	Secondary (external system)	Cambodia's national QR payment network. Receives KHQR payment requests and reports when they are paid.
ABA PayWay	Secondary (external system)	ABA Bank's payment gateway. Creates a payment (KHQR + ABA Mobile link), answers status checks and posts the result back.
Email service	Secondary (external system)	Sends the 6-digit password reset codes.

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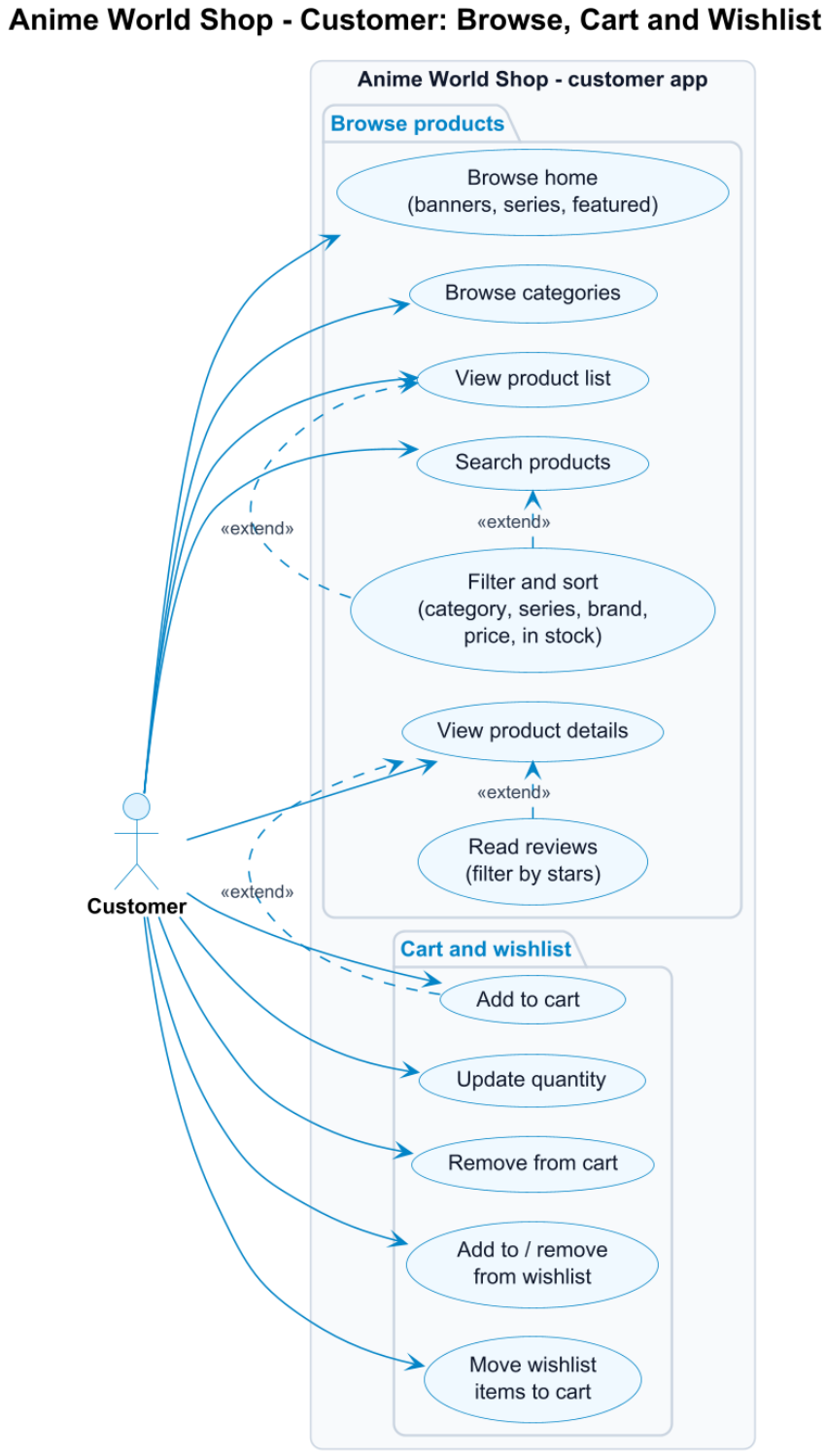
2. Use case diagrams

2.1 Overview - all actors

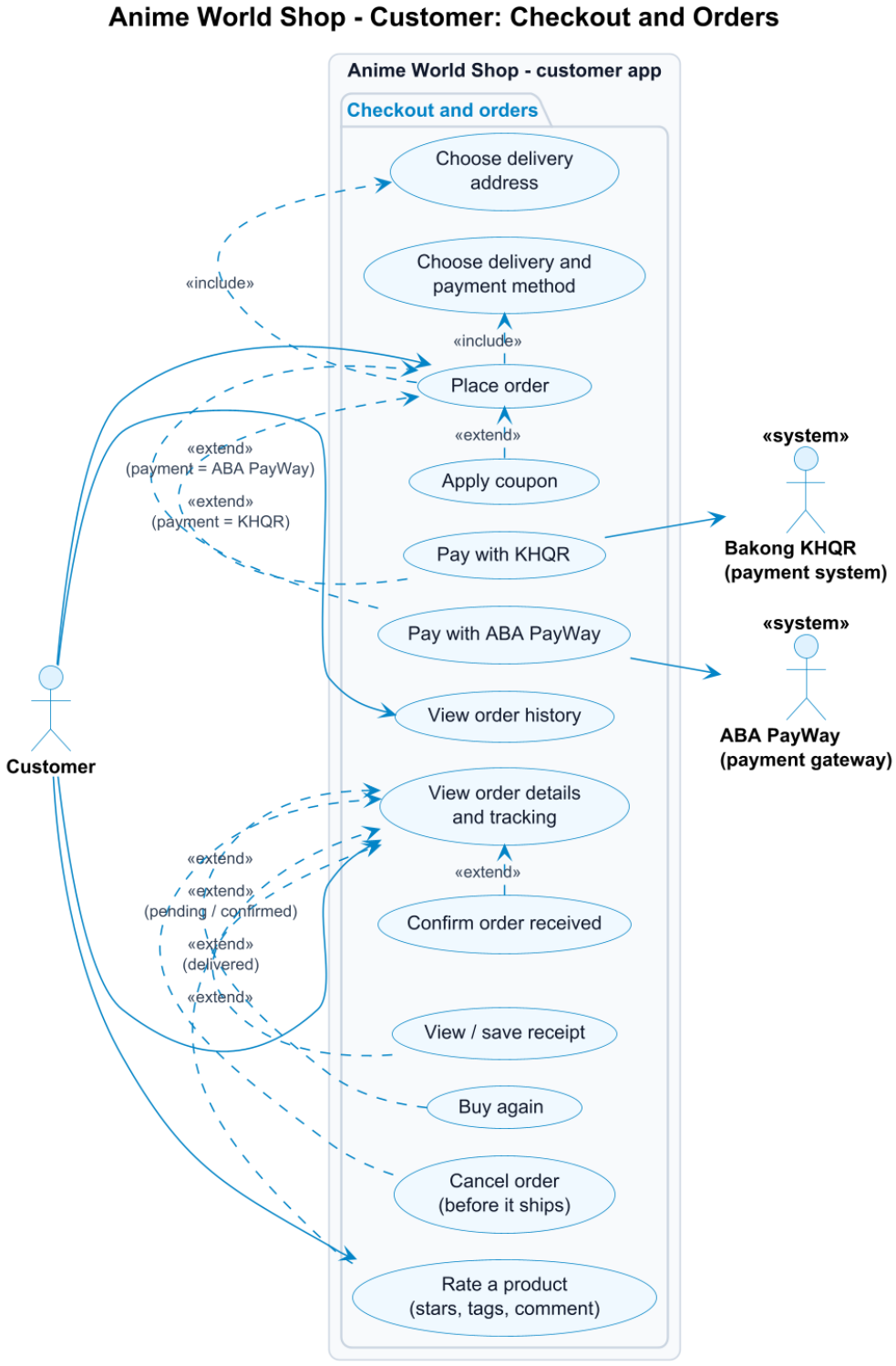


<<include>> = always part of the base use case. <<extend>> = optional extra behaviour that happens only in some cases.

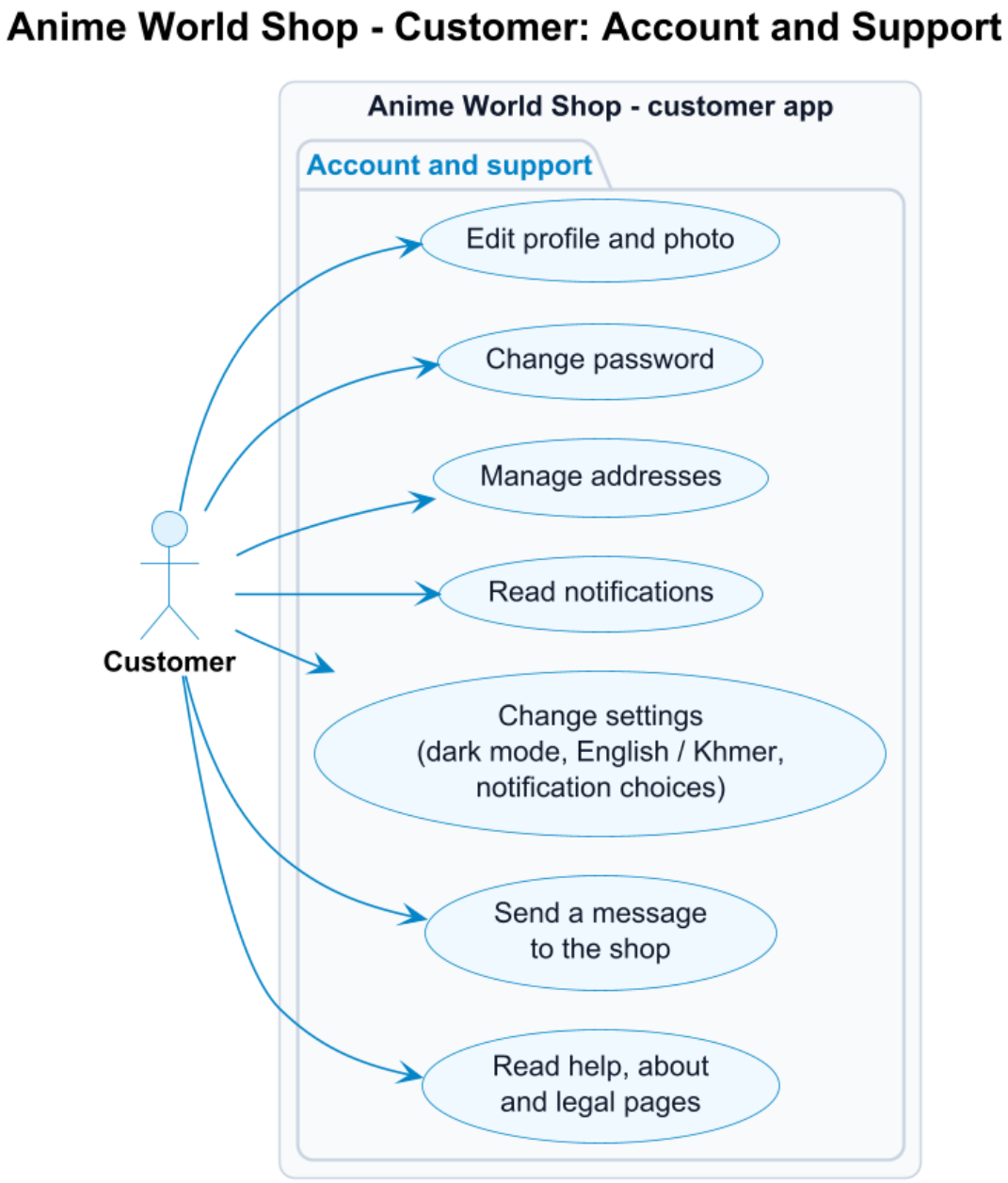
2.2 Customer - browse, cart and wishlist



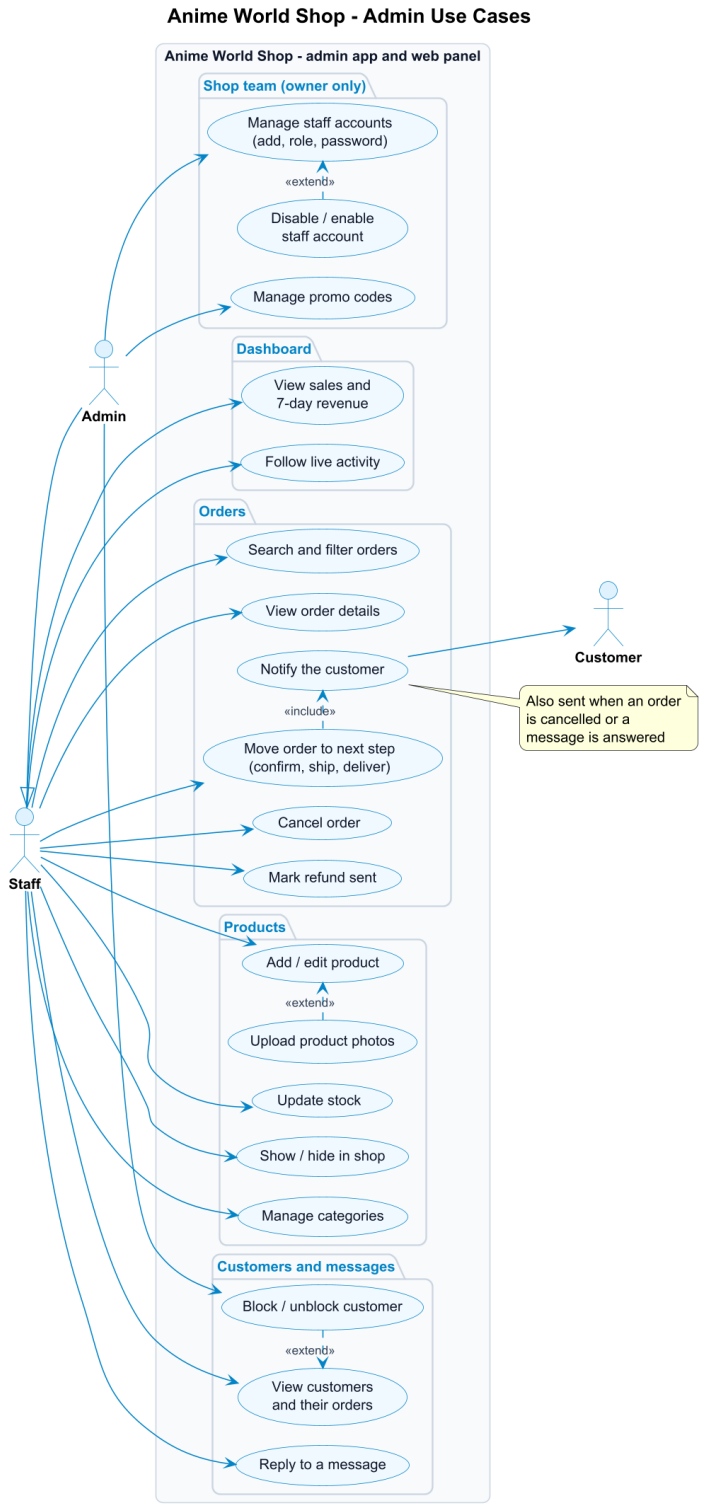
2.3 Customer - checkout and orders



2.4 Customer - account and support



2.5 Admin



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3. Use case list

ID	Use case	Actor	Requirement	Screen(s)

ID	Use case	Actor	Requirement	Screen(s)
UC-01	Register	User	FR-U02	Register
UC-02	Log in / log out	User	FR-U03, FR-U09	Login, Profile
UC-03	Reset forgotten password	User, Email service	FR-U04	Forgot Password, Verify Code, Reset Password
UC-04	Edit profile and photo	Customer	FR-U05, FR-U06	Profile, Edit Profile
UC-05	Change password	Customer, Admin	FR-U07	Change Password
UC-06	Manage addresses	Customer	FR-U08	My Addresses, Address Form
UC-07	Browse home and categories	Customer	FR-P01, FR-P02	Home, Categories
UC-08	Search, filter and sort products	Customer	FR-P05, FR-P06	Search, Product List, Filter
UC-09	View product details	Customer	FR-P04	Product Details
UC-10	Add to cart	Customer	FR-C01	Product List, Product Details
UC-11	Update quantity / remove from cart	Customer	FR-C02, FR-C03	Cart
UC-12	Manage wishlist	Customer	FR-C06	Wishlist
UC-13	Place order (checkout)	Customer	FR-O01 - FR-O03	Checkout, Order Placed
UC-14	Apply coupon	Customer	FR-C05	Cart
UC-15	Pay with KHQR	Customer, Bakong KHQR	FR-O04	KHQR Payment
UC-15b	Pay with ABA PayWay	Customer, ABA PayWay	FR-O04b	ABA PayWay Payment
UC-16	View order history and details	Customer	FR-O05, FR-O06	My Orders, Order Details
UC-17	Confirm received / buy again / view receipt	Customer	FR-O07	Order Details, Receipt
UC-17b	Cancel an order	Customer	FR-O08	Order Details
UC-18	Read notifications	Customer	FR-A01	Notifications
UC-19	Change settings	Customer, Admin	FR-A02	Settings
UC-20	Send a message to the shop	Customer	FR-A04	Contact Us
UC-21	Read help, about and legal pages	Customer	FR-A03, FR-A05	Help, About, Legal
UC-22	Rate a product (edit / delete my review)	Customer	FR-R01, FR-R02	Rate sheet (from Order Details, Notifications, To review, Product Details)
UC-23	Read reviews	Customer	FR-R03, FR-R04	Product Details, Reviews
UC-A1	View dashboard and live activity	Admin	FR-AD01	Admin Dashboard
UC-A2	Manage orders (next	Admin	FR-AD02	Admin Orders, Admin Order Details

ID	Use case	Actor	Requirement	Screen(s)
	step, cancel)			
UC-A3	Add / edit product and photos	Admin	FR-AD03	Admin Products, Add / Edit Product
UC-A4	Update stock, show / hide product	Admin	FR-AD03	Admin Products
UC-A5	Manage categories	Admin	FR-AD04	Admin Categories
UC-A6	View and block / unblock customers	Admin	FR-AD05	Admin Customers, Customer Details
UC-A7	Reply to a message	Admin	FR-AD06	Admin Messages
UC-A8	Manage staff accounts	Admin	FR-AD10, FR-AD11	Admin Staff, Add / Edit Staff Member
UC-A9	Send a refund (mark refunded)	Staff, Admin	FR-AD12	Admin Orders, Order Details, web admin Orders

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4. Use case specifications

The main use cases in detail. Numbers in the flows are steps; "A" flows are alternatives or errors that branch off a step.

UC-01 Register

Actor	User (not signed in)
Goal	Create a customer account and start shopping.
Precondition	The user has no account with this email.
Trigger	The user taps "Create account" on the Login screen.

Main flow

1. The user enters full name, email, phone (optional) and password, and ticks "I agree to the Terms & Privacy Policy".
2. The app checks the fields and sends `POST /api/auth/register`.
3. The system validates the data, stores the password as a bcrypt hash and creates the account with the role *customer*.
4. The system returns a login token and creates a welcome notification.
5. The app saves the token and opens the Home screen.

Alternative flows

- A1 (step 2/3): a field is wrong (empty name, invalid email, password shorter than 6 characters, invalid phone). The system returns `422` with the wrong fields; the app shows each message under its field.
- A2 (step 3): the email is already registered. `422`; the app shows "Email is already registered" under the email field.
- A3: no connection. The app shows "Cannot connect to server. Check your internet connection."

Postcondition A new customer account exists and the user is logged in.

UC-02 Log in

Actor	User (not signed in) - becomes Customer or Admin
Precondition	The user has an account.

Main flow

1. The user enters email and password and taps Log in.
2. The app sends `POST /api/auth/login`.
3. The system checks the password against the stored hash and that the account is active.
4. The system returns a JWT (valid 7 days) and the user's profile, including the role.
5. The app opens the shop for a customer, or the Admin Dashboard for an admin.

Alternative flows

- A1 (step 3): wrong email or password. `401`, "Incorrect email or password"; the fields shake.
- A2 (step 3): the account is blocked. `403`, "This account has been disabled".

Postcondition The token is stored on the device; the user stays logged in until they log out (UC-02 log out: confirm dialog, token deleted, Login screen shown).

UC-03 Reset forgotten password

Actors	User, Email service
Precondition	The user has an account but forgot the password.

Main flow

1. On Login the user taps "Forgot password?" and enters their email.
2. The system creates a 6-digit code valid for 10 minutes, stores only its hash, and sends it by email.
3. The user enters the code on the Verify Code screen.
4. The system checks the code and returns a short-lived reset token (15 minutes).
5. The user enters and confirms a new password.
6. The system saves the new password hash and marks the code as used.
7. The app shows "Password changed" and returns to Login.

Alternative flows

- A1 (step 2): the email is not registered. The app shows the same "code sent" message, so nobody can find out which emails have accounts.
- A2 (step 2): the user asks again within 60 seconds ("Please wait Ns before requesting a new code") or more than 5 times in an hour ("Too many code requests. Please try again in an hour."). `429`; the Resend button counts down.
- A3 (step 4): wrong code, or older than 10 minutes. `400`, "The code is incorrect or has expired".
- A4 (step 4): 5 wrong tries. `429`, "Too many wrong attempts. Please request a new code." The code is locked.

Postcondition The account has the new password; the old one no longer works.

UC-08 Search, filter and sort products

Actor	Customer

Precondition	Logged in.

Main flow

1. The customer taps the search bar on Home.
2. The app shows recent searches; as the customer types, it shows suggestions (`GET /api/search/suggestions`).
3. The customer submits a search or picks a suggestion.
4. The system returns matching active products (name, brand or series) a page at a time.
5. The customer opens Filters and chooses category, series, brands, price range and "in stock only", and a sort order.
6. The app reloads the list with those filters (`GET /api/products?q=&category_id=&brand=&min_price=&max_price=&in_stock=1&sort=`) and shows the active filters as chips.

Alternative flows

- A1 (step 4): nothing matches. The app shows an empty state ("Try removing a filter.").
- A2: the customer removes a chip, or taps "Clear all"; the list reloads without those filters.

Postcondition The customer sees the products that match.

UC-10 Add to cart

Actor	Customer
Extends	UC-09 View product details (and the product list)

Main flow

1. The customer taps "Add to cart" on a product.
2. The app sends `POST /api/cart` with the product and quantity.
3. The system checks that the product is active and in stock, then adds the line or increases its quantity.
4. The app plays the cart animation, updates the Cart badge and shows "Added to your cart" with a View cart button.

Alternative flows

- A1 (step 3): sold out (`409`, "This item is sold out") or the cart would exceed the stock (`409`, "Only N can be added for this item"). The app shows the message and nothing is added.

Postcondition The cart contains the product.

UC-13 Place order (checkout)

Actor	Customer
Includes	Choose delivery address; choose delivery and payment method
Extended by	UC-14 Apply coupon; UC-15 Pay with KHQR; UC-15b Pay with ABA PayWay
Precondition	Logged in; the cart is not empty.

Main flow

1. In the Cart the customer taps "Go to checkout".
2. The app loads `GET /api/checkout`: cart lines, addresses, delivery options and totals.

3. The customer chooses (or adds) a delivery address.
4. The customer chooses Standard delivery ($2, free from $50) or Store pickup (free).
5. The customer chooses KHQR, ABA PayWay or Cash on delivery and can add a note.
6. The customer taps "Place order"; the app sends `POST /api/orders`.
7. The system checks every line against the current stock, then in one transaction takes the stock (a conditional update, so the last item can't be sold twice), creates the order (number like `AWS-20261016-00042`) and its items, and empties the cart.
8. The system creates an "Order placed" notification.
9. Cash on delivery: the app shows the Order Placed screen. KHQR: continue with UC-15. ABA PayWay: continue with UC-15b.

Alternative flows

- A1 (step 3): no address yet. The app opens the Address Form first.
- A2 (step 7): an item sold out or has less stock than ordered since it was added. `409`, e.g. "Some items changed. Only 1 left of Nendoroid Anya Forger."; nothing is ordered and the customer can fix the cart.
- A4 (step 6): the cart is empty. `409`, "Your cart is empty".
- A3 (step 6): no payment method chosen. `422`, "Choose KHQR, ABA PayWay or Cash on delivery".

Postcondition A *pending* order exists, stock is reserved and the cart is empty.

UC-15 Pay with KHQR

Actors	Customer, Bakong KHQR
Extends	UC-13 Place order, when the payment method is KHQR
Precondition	A pending, unpaid KHQR order.

Main flow

1. The app requests a payment (`POST /api/orders/<id>/khqr`).
2. The system generates a KHQR code for the order total with the shop's Bakong account and stores its MD5 reference. The code expires after 15 minutes.
3. The app shows the QR, the amount and a countdown.
4. The customer scans the code with any Cambodian bank app (ABA, ACLEDA, Wing ...) and pays.
5. The app checks the status every few seconds (`GET /api/orders/<id>/payment`); the system asks Bakong whether that reference is paid.
6. When Bakong reports *paid*, the system marks the payment and the order as paid and sends a "Payment received" notification.
7. The app prints out the receipt and offers Save receipt, View order and Continue shopping.

Alternative flows

- A1 (step 5): 15 minutes pass without payment. The order is cancelled and its stock returned; the app shows "This QR code expired" and a Try again button.

Postcondition The order is *paid* and waits for the shop to confirm it.

UC-15b Pay with ABA PayWay

Actors	Customer, ABA PayWay

Extends	UC-13 Place order, when the payment method is ABA PayWay
Precondition	A pending, unpaid ABA PayWay order.

Main flow

1. The app requests a payment (`POST /api/orders/<id>/payway`).
2. The system sends PayWay a signed purchase (HMAC-SHA512 over the fields in PayWay's order) with the order total, the items and a 15-minute lifetime, and stores PayWay's transaction id, KHQR and ABA Mobile link.
3. The app shows the ABA PayWay card (QR, amount, countdown) and an **Open ABA Mobile** button.
4. The customer taps Open ABA Mobile and confirms the payment there, or scans the QR with any bank app.
5. The app checks the status every few seconds (`GET /api/orders/<id>/payment`); the system asks PayWay (`check-transaction-2`) about that transaction.
6. When PayWay reports *APPROVED*, the system marks the payment and the order as paid and sends a "Payment received" notification.
7. The app prints out the receipt ("Paid by ABA PayWay") and offers Save receipt, View order and Continue shopping.

Alternative flows

- A1 (step 2): PayWay refuses or cannot be reached. `502`, "ABA PayWay is not available right now"; the order stays pending and the customer can try again.
- A2 (step 4): ABA Mobile is not installed. The app says so and offers the app store; the QR still works with any bank app.
- A3 (step 5): PayWay posts the result to `/api/payments/payway/callback`. The system does not trust the posted data: it asks PayWay itself (step 5) before marking anything paid.
- A4 (step 5): 15 minutes pass without payment. As UC-15 A1: the order is cancelled and its stock returned.

Postcondition The order is *paid* and waits for the shop to confirm it.

UC-16 View order history and details

Actor	Customer

Main flow

1. The customer opens My Orders from Profile.
2. The app shows orders in progress (with an animated tracker) and past orders (`GET /api/orders`).
3. The customer opens an order; the app shows the tracker (placed, paid, shipped, delivered), the items with photos, the address, the payment and the totals.
4. Optional (UC-17): when a paid KHQR or ABA PayWay order is on the way the customer taps "Got it" to confirm it was received (the shop finishes store pickup and cash orders, because it hands them over or takes the money); for a finished order they can tap "Buy again" (the items go back into the cart); for a paid order they can view, save or share the receipt.

Alternative flows

- A1 (step 2): no orders yet. Empty state with "Start shopping".

UC-17b Cancel an order

Actor	Customer
Extends	UC-16 View order history and details
Precondition	The customer's order is pending or confirmed (not shipped yet).

Main flow

1. In Order Details the customer taps **Cancel order** (shown only while it can be cancelled).
2. The app asks "Cancel this order?" and offers optional reasons (Changed my mind, Ordered by mistake, Found a better price, It takes too long).
3. The customer picks a reason (or none) and taps **Cancel order**.
4. The system checks the order is still pending or confirmed, cancels it, puts its items back on sale and stops waiting for its payment (`POST /api/orders/<id>/cancel`).
5. If it was paid by KHQR or ABA PayWay, the payment becomes **Refund pending** and the message says the money comes back within 3 working days.
6. The customer gets a notification; the order moves to Past orders with "Cancelled" (or "Refund pending").

Alternative flows

- A1 (step 3): the customer taps "Keep my order". Nothing changes.
- A2 (step 4): the order has shipped in the meantime. `409`, "This order is already on its way, so it can't be cancelled".
- A3 (later): the shop marks the refund as sent (UC-A9). The order shows **Refunded** and the customer gets a "Refund sent" notification.

Postcondition The order is cancelled, its stock is back on sale, and any payment is owed back to the customer.

UC-22 Rate a product

Actor	Customer
Precondition	The product is in one of the customer's delivered orders.
Trigger	"Rate it" on an item of a delivered order, on the "Delivered" notification, in Profile > To review, or "You bought this. Rate it" on the product page.

Main flow

1. The app opens the rating sheet with the product photo and name ("Rate your figure").
2. The customer taps a star; the star pops and the word under it changes (Terrible ... Excellent). Post review stays disabled until a star is chosen.
3. The customer may pick quick tags and write a comment.
4. The customer taps Post review; the app sends `POST /api/products/<id>/reviews`.
5. The system checks that the product was delivered to them, validates the data and saves the review (or updates their existing one).
6. The sheet shows "Thanks for your review!" and closes; the item now shows its stars and "Reviewed", and the product's rating is updated.

Alternative flows

- A1 (step 5): not delivered to this customer. `403`, "You can review a product once it has been delivered to you".

- A2 (step 5): comment longer than 1,000 characters, unknown tag or rating not 1-5. `422`; the sheet shows the message and stays open.
- A3: the customer already reviewed it. The sheet opens as "Edit your review" with their stars, tags and comment; Save changes edits it, Delete review (after a confirmation) removes it.

Postcondition The review is on the product page and the Reviews page, marked as a verified buyer.

UC-A2 Manage orders

Actor	Admin
Includes	Notify the customer
Precondition	Logged in as admin.

Main flow

1. The admin opens the Orders tab (the badge shows how many orders need handling).
2. The app lists all orders (`GET /api/admin/orders`) with counts per status; the admin can search by order number or customer, or filter by status.
3. The admin taps the order's next-step button (Confirm, Ship or Delivered), or opens the order details first.
4. The system checks that the step is allowed (pending -> confirmed -> shipped -> delivered) and saves the new status (`PUT /api/admin/orders/<id>/status`).
5. The system creates a notification for the customer ("Your order has shipped" ...).
6. The app updates the list, the badge and the dashboard.

Alternative flows

- A1 (step 3): the admin taps Cancel. The app asks "Cancel this order?"; on confirm the order is cancelled, its stock returned and the customer notified. A paid KHQR / ABA PayWay order becomes **Refund to send** (UC-A9).
- A2 (step 4): the step is not allowed (e.g. shipping a pending order, cancelling a shipped one). `409`, "A shipped order cannot be marked cancelled".
- A3 (step 4): an unpaid KHQR order can only be cancelled, not confirmed.

Postcondition The order has its new status and the customer knows.

UC-A9 Send a refund (mark refunded)

Actor	Staff or Admin (and the Customer who gets the money back)
Precondition	A cancelled order that was paid by KHQR or ABA PayWay (payment status *refund_pending*).

Main flow

1. The shop sees the order in Orders under **Refund to send** (also a "Refund" button in Live activity).
2. The shop sends the order total back to the customer itself (bank transfer or ABA).
3. The shop opens the order and taps **Mark refunded**; the app asks to confirm that the money was sent.
4. The system marks the payment *refunded* with the date (`POST /api/admin/orders/<id>/refund`) and notifies the customer ("Refund sent").

Alternative flows

- A1 (step 4): the order has no refund waiting (already refunded, or never paid). `409`, "This order has no refund waiting".

Postcondition The customer sees **Refunded** on the order; the refund is no longer on the shop's list.

UC-A3 Add / edit product

Actor	Admin
Extended by	Upload product photos

Main flow

1. The admin taps the raised + button (or a product in the Products tab to edit it).
2. The admin fills in name, category, series, brand, price, sale price, stock, description, Featured and In the shop, and picks photos from the gallery.
3. The admin taps "Save product"; the app sends `POST /api/admin/products` (or `PUT` for an edit).
4. The system validates the data and saves the product with a URL slug.
5. The app uploads each new photo (`POST /api/admin/products/<id>/images`); the first one becomes the main photo.
6. The product appears in the shop (if "In the shop" is on).

Alternative flows

- A1 (step 4): a field is wrong (no name, price not a number, sale price not lower than the price, negative stock). `422`; the app shows each message under its field.
- A2 (step 5): a photo is not a JPG, PNG or WebP image, or is larger than 5 MB. `422`; that photo is refused with a message and the product is still saved. Accepted photos are cropped square and stored as WebP.

Postcondition The product exists with its photos.

UC-A6 View and block / unblock customers

Actor	Admin

Main flow

1. From More, the admin opens Customers and can search by name, email or phone, or filter All / Active / Blocked.
2. The admin opens a customer and sees their contact details, orders, total spent and addresses.
3. The admin taps "Block customer" and confirms.
4. The system marks the account inactive (`PUT /api/admin/customers/<id>`).
5. From now on the customer's login and any existing token are refused (`403`).

Alternative flows

- A1: the admin taps "Unblock customer"; the account is active again at once.

Postcondition The customer can, or can no longer, use the shop.

UC-A7 Reply to a message

Actors	Admin (and the Customer who receives the answer)

Includes	Notify the customer

Main flow

1. The admin opens Messages (from More or the dashboard feed); open questions are listed first.
2. The admin taps Reply, writes the answer and taps "Send reply".
3. The system saves the reply, marks the message *answered* (`POST /api/admin/contact-messages/<id>/reply`) and notifies the customer.
4. The message moves to the Answered list.

Postcondition The customer sees the answer in Notifications and on the Contact screen.

UC-A8 Manage staff accounts

Actor	Admin (shop owner)
Precondition	Logged in with an admin account. Staff accounts do not see Staff.

Main flow

1. The admin opens Staff (More > Staff in the app, or Staff in the web admin menu). Admins are listed first, then staff, with role, status and last login (`GET /api/admin/staff`).
2. The admin taps **Add staff**, enters name, email, optional phone, role (Staff or Admin) and a starting password, and saves (`POST /api/admin/staff`).
3. The system checks the fields and that the email is not registered, creates the account and lists it.
4. The admin gives the new person their email and password; they log in and get the admin area with only the staff tools.
5. Later the admin opens a person to change their name, phone or role, or taps **Reset password** to give them a new one (`PUT /api/admin/staff/<id>`). A new role or password signs them out on every device.

Alternative flows

- A1 (step 3): A field is wrong or the email is taken. `422`; the form shows what to fix.
- A2 (step 5): The admin taps **Disable** and confirms. The account is signed out at once and cannot log in until the admin taps **Enable**.
- A3 (step 5): The admin tries to change their own role or disable themselves, or to demote / disable the last active admin. `409`; nothing changes (the app already hides these buttons on your own account).

Postcondition The shop team matches who is allowed to work on the shop.